GMAT Data Insights

Two-Part Analysis

Lesson Material

Comprehensive guide covering two related questions, constraint problems, logical TPA, and efficient solving strategies.

Via G. Frua 21/6, Milano | www.uptoten.it

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EDUCATIONAL PURPOSE: This material is intended solely for educational purposes to help students prepare for standardized tests.

1. Lesson Overview

Topic: Two-Part Analysis (TPA)
Section: Data Insights
Lesson Sequence: DI-04 (Fourth of 5 DI topics)
Total Sessions: 3 lessons (A, B, C)

1.1. Learning Objectives

By the end of this topic, students should be able to:

1. Understand the TPA format with two columns and shared options
2. Identify relationships between the two parts
3. Solve mathematical TPA problems (equations, constraints)
4. Solve logical TPA problems (strengthen/weaken style)
5. Determine which part to solve first
6. Check answer consistency efficiently

1.2. GMAT Relevance

Two-Part Analysis tests the ability to solve interconnected problems—a skill essential for business analysis where multiple related factors must be considered simultaneously.

2. Part 1: TPA Question Format

2.1. Structure

Every TPA Question Has:

1. A problem statement or scenario
2. A table with two question columns and shared answer options
3. You select ONE option for each column

Both selections must come from the SAME set of answer options.

Sample TPA Format:

A store sells two products. Product A costs $15 and Product B costs $20. A customer spent exactly $100.

Product A Qty	Product B Qty	Options
○	○	0
○	○	1
○	○	2
●	○	4
○	●	2
○	○	5

Select 4 for Product A ($4 \times \$15 = \$60$) and 2 for Product B ($2 \times \$20 = \$40$)
Total $= \$100$ ✓

2.2. Scoring

Each column selection is scored independently.

- Getting one correct but not the other: partial credit
- Both must be from the same option list

2.3. Visual: TPA Answer Format

Two-Part Analysis Answer Grid

Column A	Column B	Options	
○	○	Option 1	
○	○	Option 2	
●	○	Option 3	Selected for A
○	●	Option 4	Selected for B
○	○	Option 5	

Key: Select ONE option per column — both from the same list

3. Part 2: Types of TPA Problems

3.1. Type 1: Mathematical/Constraint Problems

Characteristics:
- Two unknowns related by equations or constraints
- Answer options are numerical values
- Solutions must satisfy given conditions

Given: $x + y = 10$ **and** $x - y = 4$

Find x and y from the options.

Solving: $x = 7$, $y = 3$

3.2. Type 2: Logical/Argument Problems

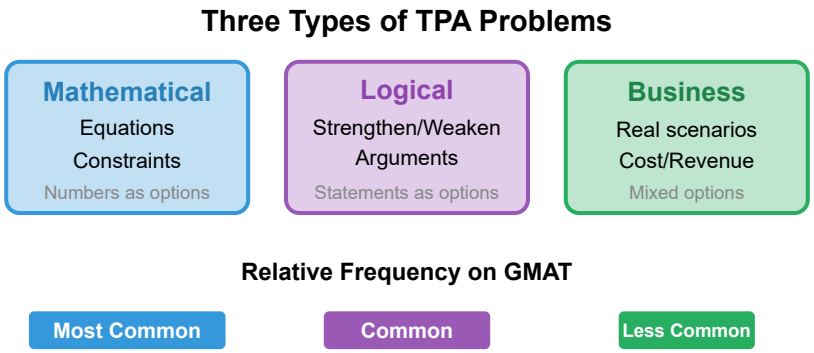
Characteristics:
- Argument or scenario described
- One column: “Strengthens the argument”
- Other column: “Weakens the argument”
- Options are statements or facts

These are like Critical Reasoning questions with a twist.

3.3. Type 3: Business Scenario Problems

Characteristics:
- Real-world business scenario
- Find two related quantities (cost and revenue, supply and demand)
- May involve optimization

3.4. Visual: TPA Problem Types Overview



4. Part 3: Solving Mathematical TPA

4.1. Identifying Relationships

Look for:
- Equations relating the two unknowns
- Constraints limiting possible values
- Whether the two parts are independent or dependent

4.2. Constraint Analysis

Problem: A vendor sells apples for $2 and bananas for $3. Total revenue was $50.

Constraint: $2A + 3B = 50$ (where A and B are positive integers)

Test options:
- If $A = 4$: $2(4) + 3B = 50 \rightarrow 3B = 42 \rightarrow B = 14$ ✓
- If $A = 10$: $2(10) + 3B = 50 \rightarrow 3B = 30 \rightarrow B = 10$ ✓

Multiple solutions possible! Need additional constraint or check options.

4.3. Systems of Equations

For two equations with two unknowns:
1. Use substitution or elimination
2. Verify solution satisfies both equations
3. Match to answer options

5. Part 4: Solving Logical TPA

5.1. The Strengthen/Weaken Format

Common logical TPA format:

"Select the statement that strengthens the conclusion and the statement that weakens the conclusion."

Apply Critical Reasoning skills to both columns.

5.2. Approach for Logical TPA

1. Identify the conclusion in the argument
2. Identify the evidence supporting it
3. Find the assumption (gap between evidence and conclusion)
4. Look for options that:
   - Support the assumption (strengthen)
   - Attack the assumption (weaken)

5.3. Common Logical Relationships

Strengthening:
- Provides additional evidence
- Rules out alternative explanations
- Supports the assumption

Weakening:
- Provides counter-evidence
- Suggests alternative explanations
- Attacks the assumption

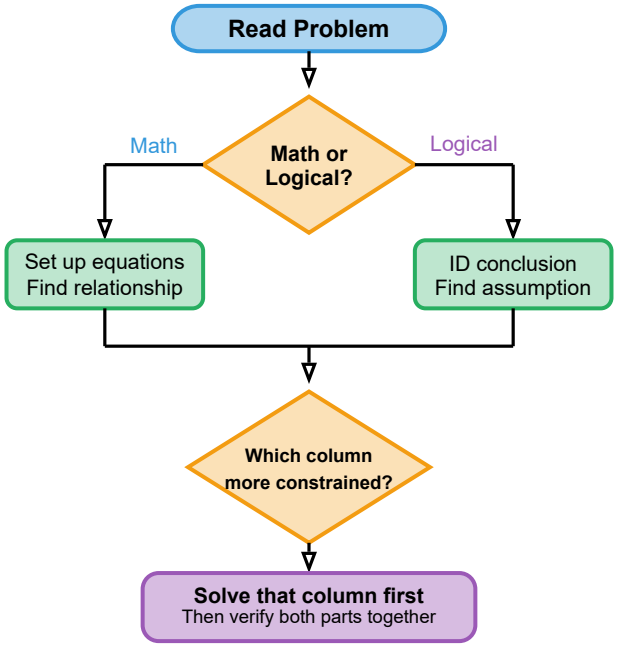
6. Part 5: Strategic Solving

6.1. Which Part First?

Solve the more constrained part first:

- If one column has fewer valid options, start there
- If one part is independent, solve it first
- If neither is clearly easier, just pick one and test

6.2. TPA Solving Flowchart



6.3. Testing Efficiency

Smart Testing:

1. Don't test all options for both columns
2. Narrow down one column first
3. Then test remaining candidates in the other

6.4. Checking Consistency

Before finalizing:

- Verify both selections satisfy ALL given conditions
- For math problems: plug values back into equations
- For logical problems: ensure one truly strengthens and one truly weakens

7. Part 6: Common TPA Patterns

7.1. Pattern 1: Sum and Difference

If given $x + y$ and $x - y$, you can find x and y:

$$x = \frac{\text{sum} + \text{difference}}{2}$$

$$y = \frac{\text{sum} - \text{difference}}{2}$$

7.2. Pattern 2: Price $\times$ Quantity

$$\text{Total} = \text{Price}_1 \times \text{Quantity}_1 + \text{Price}_2 \times \text{Quantity}_2$$

If you know the total and prices, test quantity combinations from options.

7.3. Pattern 3: Ratio Problems

If items are in ratio $a : b$ and total is known:

- First item $= \dfrac{a}{a+b} \times \text{Total}$
- Second item $= \dfrac{b}{a+b} \times \text{Total}$

7.4. Pattern 4: Rate $\times$ Time Problems

$\text{Distance} = \text{Rate} \times \text{Time}$ or $\text{Work} = \text{Rate} \times \text{Time}$

7.5. Visual: Common Patterns at a Glance

Quick Reference: TPA Pattern Formulas

Sum & Difference

$$x = \frac{S+D}{2}, \quad y = \frac{S-D}{2}$$

Price $\times$ Quantity

$$P_1Q_1 + P_2Q_2 = \text{Total}$$

Ratio $a : b$

$$\text{Part} = \frac{a}{a+b} \times \text{Total}$$

Rate $\times$ Time

$$D = R \times T, \quad W = R \times T$$

Tip: Memorize these patterns to save time!
Recognize the pattern → Apply the formula → Test options

8. Part 7: Time Management

8.1. Pacing Guidelines

Target Time: 2-2.5 minutes per TPA question

TPA requires careful analysis but shouldn't take too long if you identify the relationship quickly.

8.2. Efficiency Tips

1. **Identify problem type immediately** - math or logical?
2. **Write down relationships/constraints** - don't solve in your head
3. **Eliminate impossible options** - reduce testing
4. **Check both parts together** - they must be consistent

9. Teaching Notes for Tutors

9.1. Lesson A Focus (Introduction)

Prioritize:

1. TPA format and two-column structure
2. Simple mathematical TPA (two equations)
3. Basic constraint problems
4. Testing approach

Question Time: 5-6 mathematical TPA questions

9.2. Lesson B Focus (Deep Dive)

Prioritize:

1. Logical/argument TPA
2. Complex business scenarios
3. Efficiency strategies
4. Common patterns and shortcuts

Review errors from Training 1, focusing on:

- Not checking consistency between parts
- Inefficient testing strategies
- Misidentifying logical relationships

9.3. Lesson C Focus (Assessment Prep)

Brief review of:

- Any patterns from Training 2 errors
- Quick pattern recognition
- Time management strategies

Assessment: 20 questions, 50 minutes

9.4. Common Student Difficulties

1. Not recognizing the relationship between parts
2. Testing too many options inefficiently
3. Choosing inconsistent answers
4. Struggling with logical TPA vs. math TPA
5. Spending too long on complex calculations

Tutor Tip: Have students explicitly state the relationship between parts before solving. This prevents inconsistent answers.